

PHASE 3 – PROJECT DESIGN PHASE

3.1 Overview

The **Project Design Phase** focuses on transforming the requirements into a detailed blueprint for the **Medical Inventory Management System**.

It defines how the system will be structured on the **Salesforce platform**, including the design of custom objects, relationships, workflows, automation, and user interfaces.

This phase ensures that the system design aligns with business goals, supports scalability, and delivers a smooth user experience.

3.2 System Architecture

The system follows a **multi-layered Salesforce architecture**:

- **Presentation Layer:** Salesforce Lightning Experience (User Interface) where users interact with the application.
 - **Application Layer:** Business logic implemented using **Apex Classes, Triggers, and Flows**.
 - **Data Layer:** Custom and standard Salesforce objects that store all inventory, supplier, and transaction data.
 - **Integration Layer (Optional):** Can connect with external billing or hospital systems via **Salesforce APIs**.
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3.3 Design Components

a) Custom Objects

1. **Medicine / Equipment**
 - Fields: Item Name, Batch Number, Quantity, Expiry Date, Reorder Level, Unit Price, Category.
 2. **Supplier**
 - Fields: Supplier Name, Contact Number, Email, Address, Supplied Items.
 3. **Purchase Order**
 - Fields: Order ID, Supplier (Lookup), Ordered Items, Quantity, Order Date, Delivery Date, Status.
 4. **Inventory Transaction**
 - Fields: Transaction ID, Item (Lookup), Transaction Type (Issue/Receive), Quantity, Date, Remarks.
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b) Relationships

- **Supplier → Medicine:** One-to-Many relationship (one supplier can provide many medicines).
 - **Medicine → Inventory Transaction:** One-to-Many relationship (one medicine can have many stock movements).
 - **Purchase Order → Supplier:** Many-to-One (multiple orders can belong to one supplier).
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3.4 Process Flow Design

1. Stock Management Flow

- User adds or updates stock.
- System automatically recalculates total quantity.
- Low-stock notification is triggered if below reorder level.

2. Expiry Tracking Flow

- Scheduled Flow checks expiry dates daily.
- Sends alerts for items expiring within 30 days.

3. Purchase Order Flow

- User creates a purchase request.
- Flow sends for approval.
- Upon approval, system updates stock once order is received.

4. Report & Dashboard Flow

- Real-time reports generated for stock value, expiry items, and supplier performance.
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3.5 User Interface Design

- **Lightning Record Pages** customized for each object (Medicine, Supplier, Purchase Order).
 - **Tabs and Navigation** for easy access to modules.
 - **Lightning Components / Dynamic Forms** to display item details, alerts, and related lists.
 - **Dashboards** for visual tracking of stock and supplier data.
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3.6 Automation Design

- **Record-Triggered Flows:** Auto-update quantity after new transactions.
- **Email Alerts:** Notify users for low stock or expired items.

- **Validation Rules:** Prevent saving records with invalid data (e.g., past expiry date or negative quantity).
 - **Approval Process:** For confirming large purchase orders.
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3.7 Security Design

- Define **Profiles** for Admin, Pharmacist, and Inventory Manager.
 - Use **Permission Sets** for additional access.
 - Apply **Sharing Rules** and **Role Hierarchies** to protect sensitive data.
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3.8 Output Design

- Interactive **Dashboards** showing:
 - Total Stock
 - Expired Items
 - Low-Stock Alerts
 - Supplier Performance
 - Printable **Reports** for monthly usage and reorder lists.
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3.9 Design Validation

- Conduct design walkthroughs with stakeholders.
- Review object relationships, automation, and UI flow for accuracy.
- Make adjustments before moving to development and testing phases.